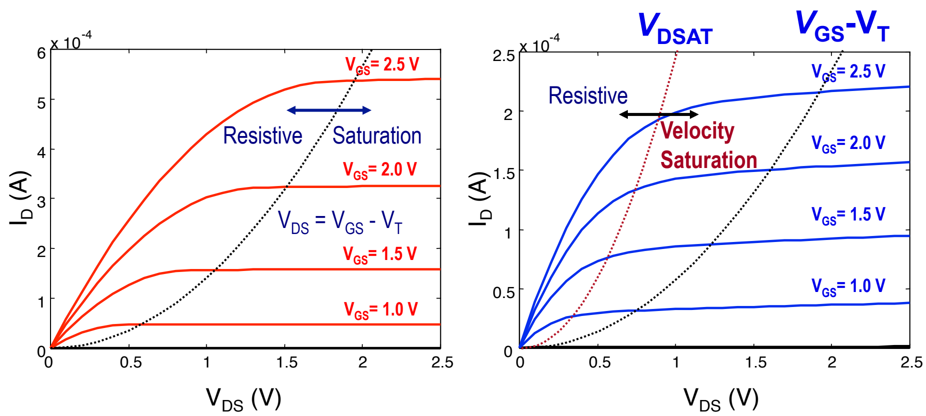
Regions of Operation



Long Channel (L=2.5μm)

W/L=1.5

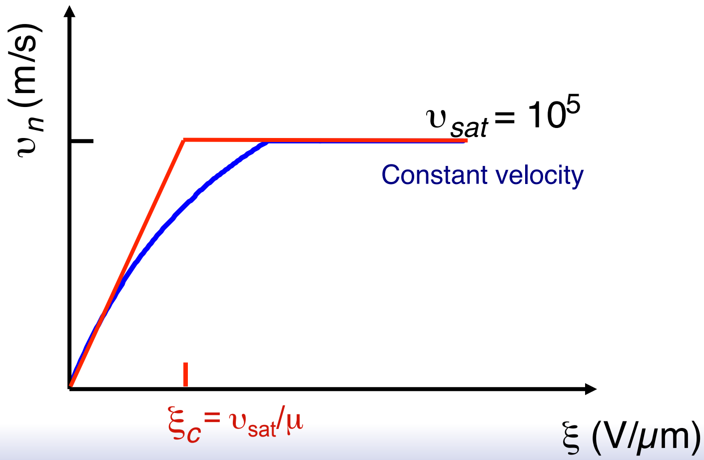
Short Channel (L=0.25μm)

Models, Models, Models…

- Exact behavior of transistor in velocity sat. somewhat challenging if want simple/easy to use models
- So, many different models developed over the years
  - v-sat, alpha, unified, V_T*, etc.
- Simple model for manual analysis desirable
  - Assume velocity perfectly linear until υ_{sat}
  - Assume V_{DSAT} constant

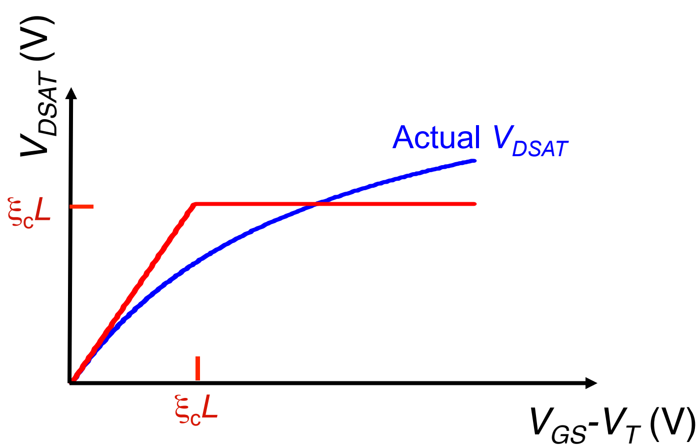
Simplified Velocity Saturation

- Assume velocity perfectly linear until hit υ_{sat}

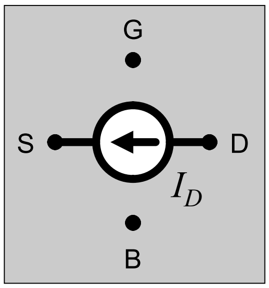


Simplified Velocity Saturation (cont'd)

- Assume $V_{DSAT} = \xi_c L$ when $(V_{GS} - V_T) > \xi_c L$



A Unified Model for Manual Analysis



define $V_{GT} = V_{GS} - V_T$

for $V_{GT} \leq 0$: $I_D = 0$

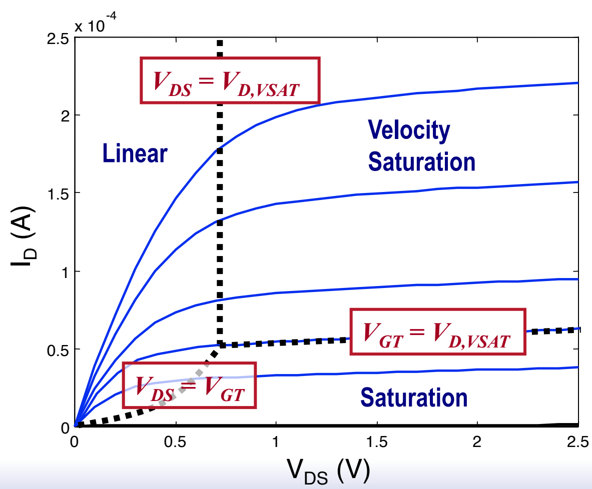
for $V_{GT} \geq 0$:

$$I_D = k' \cdot \frac{W}{L} \cdot \left(V_{GT} \cdot V_{DS,eff} - \frac{V_{DS,eff}^2}{2} \right) \cdot \left(1 + \lambda \cdot V_{DS}\right)$$

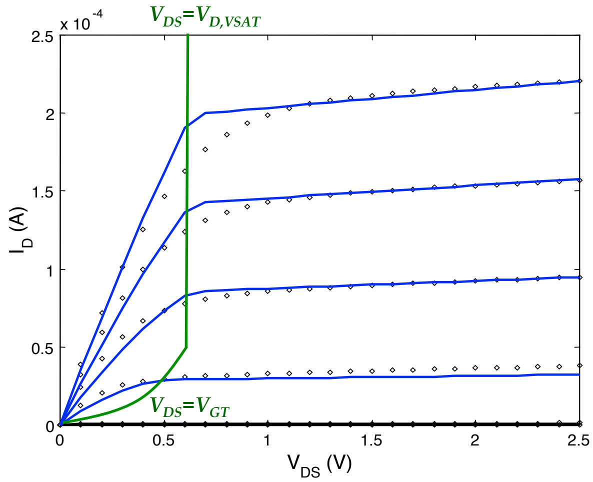
with $V_{DS,eff} = \min(V_{GT}, V_{DS}, V_{D,VSAT})$

Simplified Model

- Define $V_{GT} = V_{GS} - V_T$, $V_{D,VSAT} = \xi_c \cdot L$



Simple Model versus SPICE



One Last Simplification

- If device always operates in velocity sat.:

$$I_D = k' \cdot \frac{W}{L} \cdot \left(V_{GS} - V_T - \frac{V_{D,VSAT}}{2} \right) V_{D,VSAT}$$

- "V_T*" model:

$$V_T^* \equiv V_T + \frac{V_{D,VSAT}}{2}$$

$$I_D = k' \cdot \frac{W}{L} \cdot \left(V_{GS} - V_T^* \right) V_{D,VSAT}$$

- Good for first cut, simple analysis

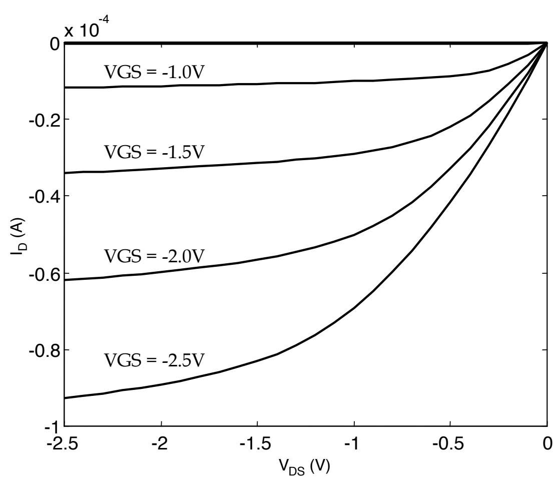
Transistor Model for Manual Analysis

Table 3.2 Parameters for manual model of generic 0.25 μm CMOS process (minimum length device).

	V_{T0} (V)	γ ($V^{0.5}$)	V_{VSAT} (V)	k' (A/V^2)	λ (V^{-1})
NMOS	0.43	0.4	0.63	115×10^{-6}	0.06
PMOS	−0.4	-0.4	-1	-30×10^{-6}	-0.1

Textbook: page 103

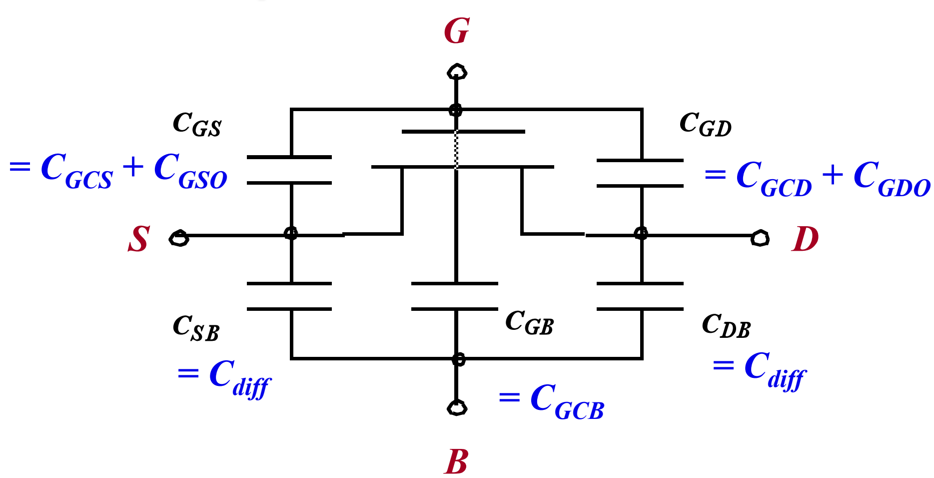
A PMOS Transistor



- All variables negative
- I prefer to work with absolute values – makes life easier.

MOS Capacitance

MOS Capacitances



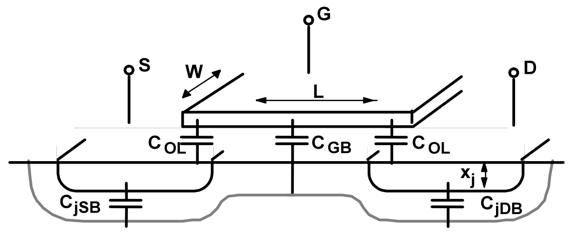
Gate Capacitance

- Capacitance (per area) from gate across the oxide is $W \cdot L \cdot C_{ox}$, where $C_{ox}=\varepsilon_{ox}/t_{ox}$

Gate Capacitance

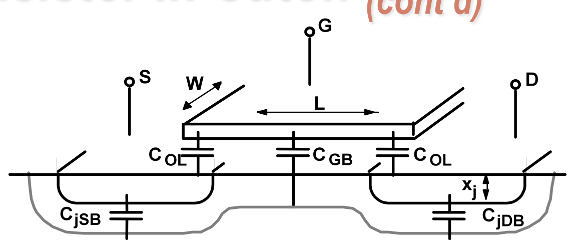
- Distribution between terminals is complex
  - Capacitance is really distributed
    - Useful models lump it to the terminals
  - Several operating regions:
    - Way off, off, transistor linear, transistor saturated

Transistor In Cutoff



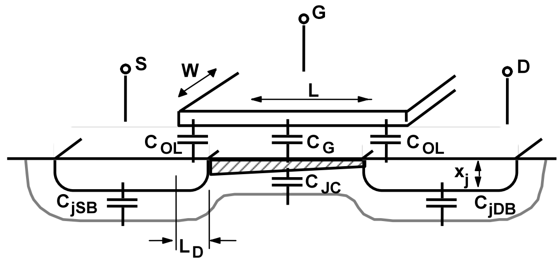
- When the transistor is off, no carriers in channel to form the other side of the capacitor.
  - Substrate acts as the other capacitor terminal
  - Capacitance becomes series combination of gate oxide and depletion capacitance

Transistor In Cutoff (cont'd)



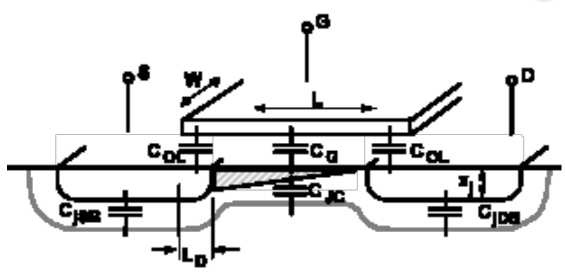
- When $|V_{GS}| < |V_T|$, total C_{GCB} much smaller than $W \cdot L \cdot C_{ox}$
  - Usually just approximate with $C_{GCB} = 0$ in this region.
- (If V_{GS} is "very" negative (for NMOS), depletion region shrinks and C_{GCB} goes back to $\sim W \cdot L \cdot C_{ox}$)

Transistor in Linear Region



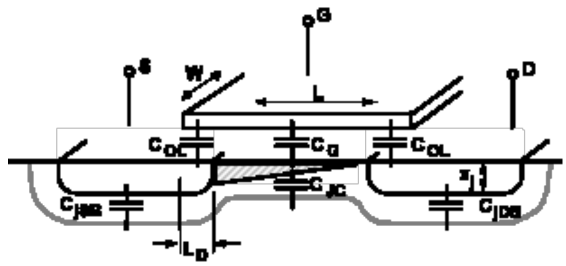
- Channel is formed and acts as the other terminal
  - C_{GCB} drops to zero (shielded by channel)
- Model by splitting oxide cap equally between source and drain
  - Changing either voltage changes the channel charge

Transistor in Saturation Region



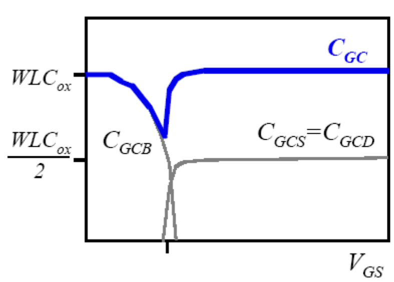
- Changing source voltage doesn't change V_{GC} uniformly
  - E.g. V_{GC} at pinch off point still V_{TH}
- Bottom line: $C_{GCS} \approx 2/3 \cdot W \cdot L \cdot C_{ox}$

Transistor in Saturation Region (cont'd)

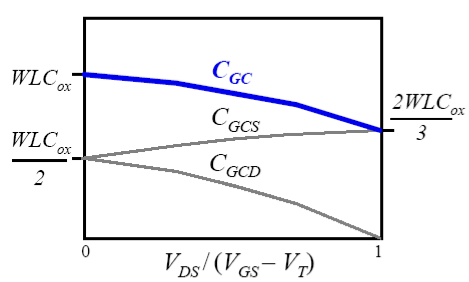


- Drain voltage no longer affects channel charge
  - Set by source and V_{DS_sat}
- If change in charge is 0, $C_{GCD} = 0$

Gate Capacitance

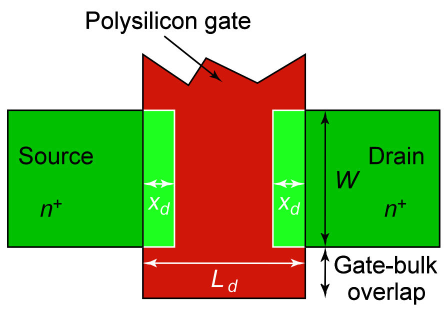


C_{gate} vs. V_{GS} (with $V_{DS} = 0$)

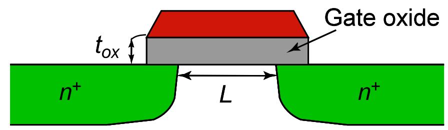


C_{gate} vs. operating region

Gate Overlap Capacitance



Top view

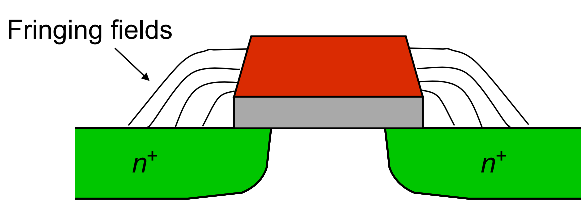


Cross section

$$C_O = C_{ox} \cdot x_d$$

Off/Lin/Sat → $C_{GSO} = C_{GDO} = C_O \cdot W$

Gate Fringe Capacitance



Cross section

- C_{OV} not just from metallurgic overlap – get fringing fields too
- Typical value: ~0.2fF·W(in µm)/edge